

Vision-Based Navigation for AMRs in a Dynamic Warehouse

Progress review

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The question

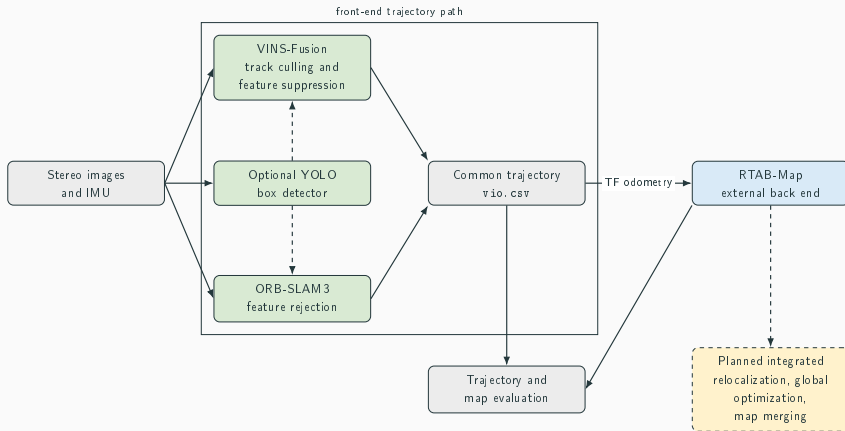
Can a vision-based pipeline provide reliable localization for an autonomous mobile robot in a warehouse *that contains moving objects*?

Two sub-questions:

1. How do the two leading stereo-inertial pipelines — ORB-SLAM3 and VINS-Fusion — behave under controlled changes in scene dynamics, floor texture and compute load?
2. Does masking dynamic objects with a learned detector improve localization, and by how much?

Path planning is outside the scope. The deliverable is a reliable navigation state and map, not a planner.

What exists at the cutoff



Green: built here. Blue: external. Dashed yellow: future work. There is deliberately *no* feedback arrow from RTAB-Map into either front end.

The methodological core: admissibility before results

Every comparison is tested for validity *before* its numbers are read. A run is descriptive unless it passes:

- same recorded input to both treatments, within 5 % on frame count and duration;
- the intended treatment demonstrably active, not merely requested;
- a common ground-truth interval, with the matched sample count reported;
- physical plausibility — no step implying more than 10 m s^{-1} .

This protocol is what produced most of the findings in this review, including two that would otherwise have entered the thesis as results.

Finding 1: the dead-reckoning floor is higher than expected

Five recordings, four worlds, all five estimators on identical data. Aligned ATE RMSE in metres.

Dataset	IMU only	Wheel	VINS	ORB	EKF
allsensor_000	21.70	2.80	0.62	0.58	0.19
allsensor_001	12.62	0.98	0.38	0.37	0.31
allsensor_002	13.24	1.49	0.56	0.60	0.26
allsensor_003	45.56	3.19	0.78	6.48	0.42
allsensor_004	26.71	3.24	0.43	1.24	0.49

A wheel+IMU EKF beats both visual pipelines on four of five datasets, at 0.10 % to 0.88 % drift. **The project finally has a proprioceptive baseline.**

...but the two families were never tuned alike

- The EKF's noise model is *measured from each recording*.
- VINS and ORB read *four constants from a hand-set file*.
- Tuning was an uncontrolled variable, confounded with the estimator.

Give the EKF that same file and re-run. Aligned ATE RMSE, metres:

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EKF, as tuned	0.19	0.31	0.26	0.42	0.49
EKF, VSLAM config	1.31	0.68	0.87	1.23	1.58
VINS-Fusion	0.62	0.38	0.56	0.78	0.43

VINS wins on five of five. The ranking is a tuning result, not an architecture result.

Why that finding matters

It is the missing baseline — plus an honest account of what it is worth.

- Before this, every trajectory figure reported visual error *with nothing to compare it to*.
- Four hypotheses were registered before the metrics were computed; one was falsified by its own data. A fifth, added after, asked whether the EKF's lead survived matched tuning. It did not.
- An ablation localises the whole effect to one term: the gyroscope process noise.

The honest reading: **the noise model is worth more than the architecture over this range**. Vision must still justify itself where dead reckoning fails.

Finding 2: the detector's limit is detection, not semantics

9860 camera frames, ground truth projected from simulator object poses.

Bounding-box detection

precision **0.684**

recall **0.593**

moving-object recall **0.591**

Class, once localized

accuracy **0.996**

37 061 associated objects

137 substitutions

The model names objects almost perfectly once it finds them, and misses about 41 % of the moving ones. Improving the classifier would change nothing; **recall is the binding constraint**.

Also: the detector outputs boxes, not segmentation, and class membership is not evidence of motion. There is no 3-D motion gate.

Finding 3: the reconstructed map is geometrically sound

Tiny warehouse, RTAB-Map database scored against the simulator's own collision geometry at 0.25 m tolerance.

Occupied-cell precision	96.6 %
Recall, region observed	89.5 %
Recall, whole building	67.2 %

The gap between 89.5 and 67.2 % is route coverage, not mapping failure: the run never entered the side aisles.

The drivability half of this evaluation was found to be scored against the run that *built* the map. Caught by comparing node poses against ground truth at matching timestamps; reported as queried rather than used.

Finding 4: neither pipeline runs at input rate

Frames arrive every 33 ms. Median per-stage cost:

	Without detector	With detector
ORB-SLAM3 tracking	63 ms	113 ms
ORB-SLAM3 local mapping	190 ms	356 ms
VINS-Fusion back end	60 ms	—
Input retained	89 %	67 %

ORB-SLAM3 discards frames at its own subscriber under load; VINS-Fusion buffers them. **The two differ in queue policy as well as algorithm**, so throughput is not yet a fair comparison between them.

Withheld: the dynamic-scene results

The dynamic worlds are **not yet a physical simulation**. Props follow scripted kinematic paths: collision geometry, but no dynamic response against the robot and no constraint to behave as warehouse traffic would.

48 runs logged, all 48 fail the plausibility check — and they are **withheld, not reported as findings**. They describe the simulation, not the problem.

A scoping decision, not a failure to measure — the runs stay in the record as excluded evidence. Two consequences: the condition this thesis is named for has **no valid testbed yet**, untested rather than tested and failed; and the environment becomes the leading candidate cause of the breakdown, as well as the cheapest to test.

What is *not* established: the core hypothesis

Semantic filtering — the contribution the thesis is named for — has **no valid test behind it yet**. Two instrumentation faults:

- **The VINS filter never fired.** All 12 runs requested it; all 12 record zero matched detector frames, zero mask coverage, zero removed tracks. The detector works offline — the masks never reached the tracker.
- **ORB pairs were not paired.** 11 of 12 baseline/filtered pairs received frame counts differing by 10 % to 30 % on the same recording, while durations matched to 0.13 %.

Both were caught by the admissibility protocol, **before** any filter-versus-baseline number was reported. One valid pair exists out of twenty-four.

These are faults in the code path, so they stand regardless of the scene. But the filtered

The audit is the result

Four claims were retracted or prevented this period:

1. A camera-rotation parsing bug that made the map a smear — found by map distortion, not by ATE, which looked plausible throughout.
2. A map judged “not navigable” — the grid had been mis-seated, so real structure was scored against the wrong cells.
3. A drivability score measured against the map’s own construction run.
4. The two filter faults above.

Every one of these would have entered the thesis as a result. The protocol that catches them is, at this stage, the most transferable output of the work.

Next period

Priority	Action	Unblocks
1	Rebuild the dynamic worlds with physically simulated props	Gives the thesis's titular co a valid testbed. Also the cl candidate cause for the bre
2	Fix the VINS mask delivery path	The core hypothesis, on th side. Needs no new data.
3	Equalise ORB input across paired runs	Makes the filter comparison sible.
4	Add disturbance to the simulation	Wheel slip and encoder qu tion; both idealisations cu favour the dead-reckoning b
5	Independent route for map drivability	One new recording, not a ne
6	Large-warehouse map evaluation	Extends map results beyond t world.

Where I need the committee's guidance

1. **Scope.** The static-scene comparison reduces to a tuning difference, and neither visual pipeline has ever been tuned to this platform. Equalise that first — tune both and re-run — or settle the question where dead reckoning is expected to fail regardless?
2. **Depth versus breadth.** Rebuild the dynamic environment properly before measuring anything further in it, or broaden to the real robot and the large warehouse with the static results as they stand?
3. **The integration direction.** Relocalization, global optimization and map merging are listed as future work. Is that the right next contribution, or a distraction until dynamic-scene tracking is sound?

Questions

Backup: static-scene repeatability

Three static datasets, three repeats each, median (IQR).

	Translational ATE	Rotational RMSE
VINS-Fusion	0.370 m (0.063)	1.56° (0.98)
ORB-SLAM3	0.454 m (0.157)	17.92° (1.72)

Positional error is comparable; rotational error differs elevenfold. But jump-free rotational RPE is 0.10° to 0.66° for ORB-SLAM3 against 0.49° to 0.69° for VINS-Fusion. **The entire gap comes from one or two discrete events per run**, not from worse orientation tracking.

Backup: evidence base

- 66 logged estimator runs across seven simulation datasets, three repeats per condition.
- Five multi-sensor recordings spanning four worlds, used for the baseline comparison.
- 9860 camera frames with projected object ground truth.
- One RTAB-Map database with graph, geometry and drivability evaluation.
- Per-frame timing, delivery, CPU and memory records for every run.

Full detail, including all retracted and excluded measurements, is in the accompanying progress report.